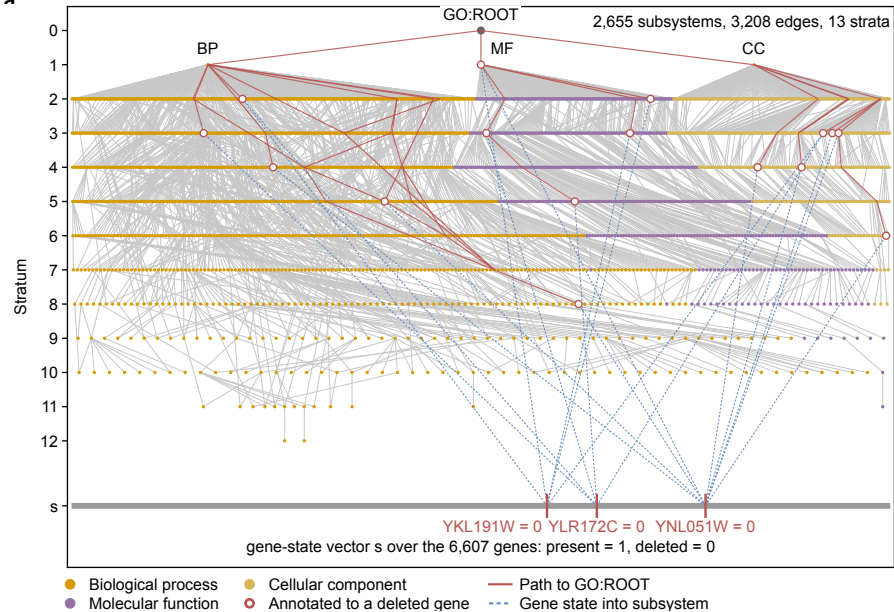


a**Perturbation enters as data**

$$s_i = 0 \text{ if } g_i \in p, \text{ else } 1$$

(term, gene) rows copied per strain; rows of deleted genes zeroed

Subsystem t , one per GO term

$$I_t = \left[\|_{c \in \text{ch}(t)} O_c \| s_{\text{genes}(t)} \right]$$

$$O_t = \tanh \left(\text{BN}(W_t I_t + b_t) \right) \in [-1, 1]^{L_t}$$

$$L_t = \max \left(20, \lceil 0.3 |\text{genes}(t)| \rceil \right)$$

Root readout: trigenic interaction

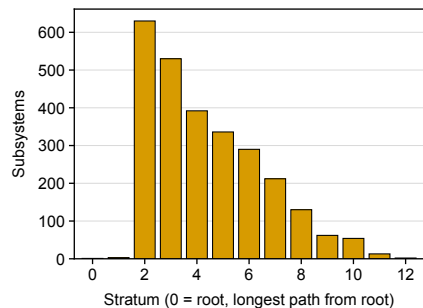
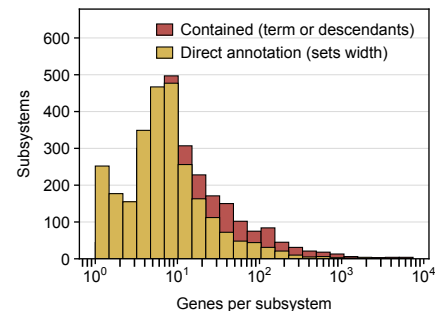
$$\hat{y} = w_r O_{\text{ROOT}} + b_r$$

Auxiliary head on every $t \neq r$

$$\hat{y}_t = w'_t O_t + b_t$$

Loss

$$\mathcal{L} = \text{MSE}(\hat{y}, y) + \alpha \text{ mean}_{t \neq r} \text{MSE}(\hat{y}_t, y), \quad \alpha = 0.3$$

b**c****d**